



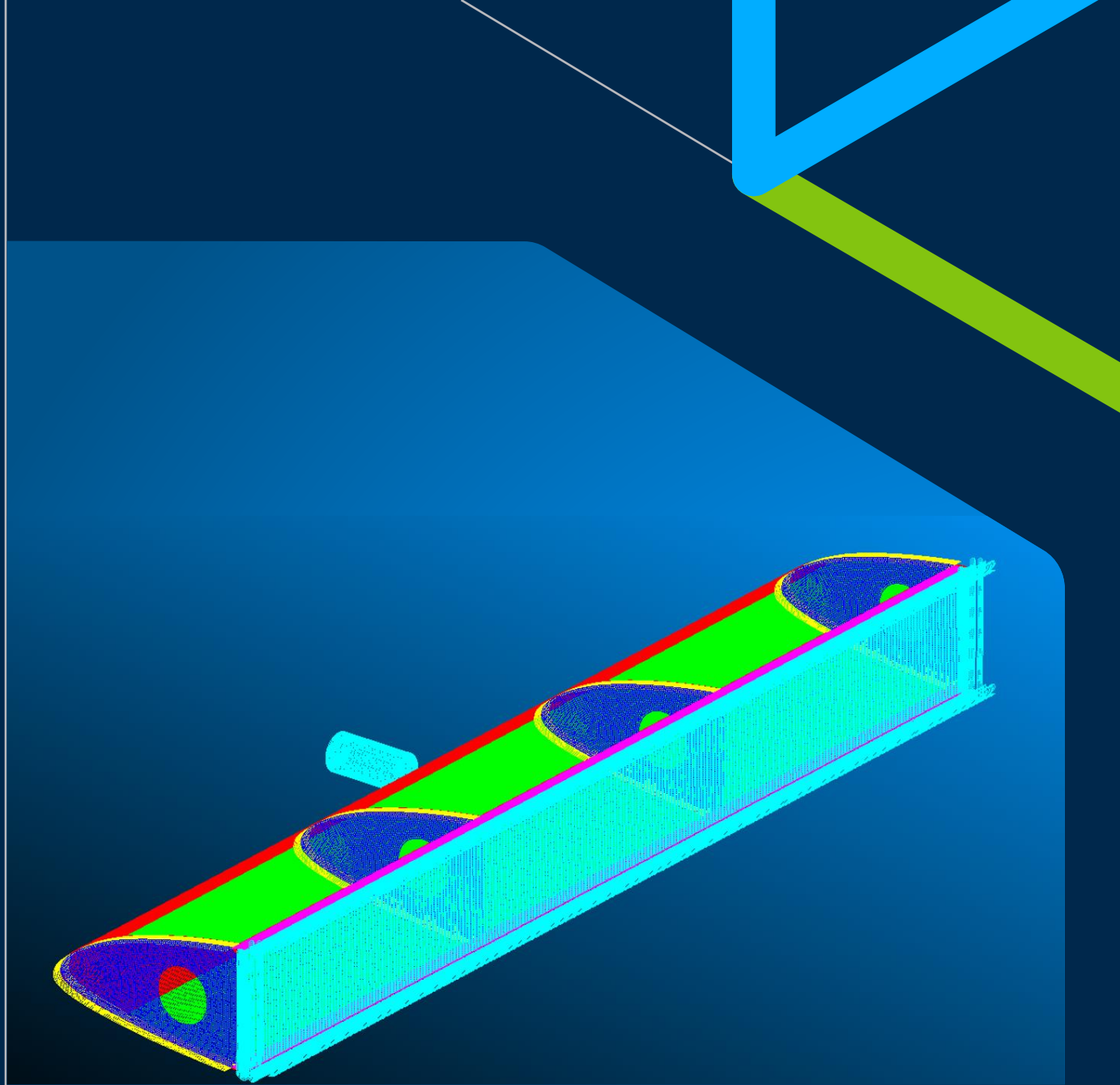
Dytran



Release 2025.2:

What's new

– October 2025



Introduction to Dytran

Dytran | Short-duration event simulation

Dytran offers three-dimensional analysis for dynamic, nonlinear behaviour of solid components, structures, and fluids. It uses explicit time integration.

Dytran is particularly suited to analyzing short, transient dynamic events that involve large deformations, a high degree of nonlinearity, and interactions between fluids and structures



The value proposition:
An explicit solution for transient structural dynamics and fluid-structure interaction



Release summary

Dytran 2025.2

Product improvements and enhancements

Product improvements in **Dytran 2025.2**

Usability enhancements for output requests

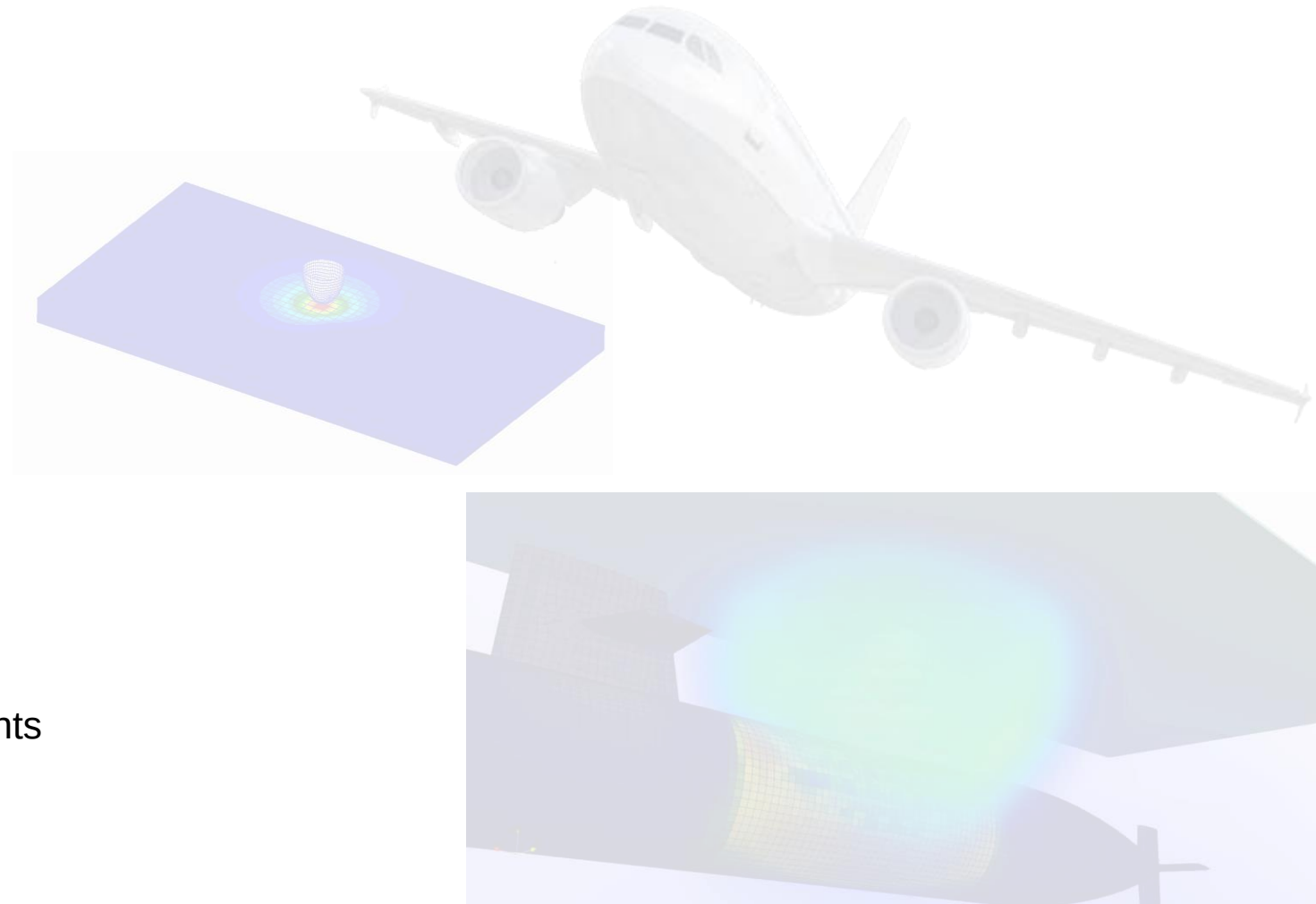
- Time history output in user-defined coordinate systems
- Center of gravity calculation for Eulerian elements

Material enhancement

- New strain-based composite failure model

Contact enhancement

- Rigid body contact performance improvements



New capabilities and enhancements in **Dytran 2025.2**

Time history output in user-defined coordinate systems

Description:

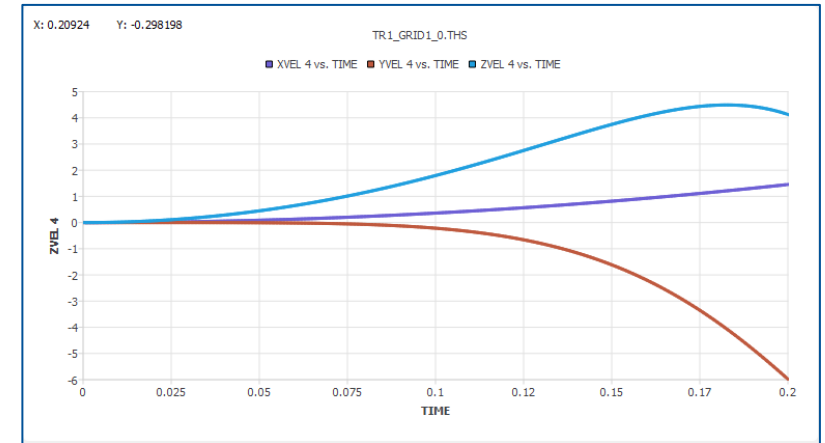
Dytran can generate time-history files during simulation that capture results at specific grid points and elements. These results may include, for instance, translations, velocities, accelerations, and forces at specific nodes or at specific rigid bodies. Using Dytran Explorer and the Visual Time History Viewer, users can efficiently plot these results or combine multiple datasets for further analysis. Up to the version 2025.1, time-history output was limited to the global coordinate system. Starting with this release, local coordinate systems—either stationary or moving—can now be selected for result generation.

Benefits:

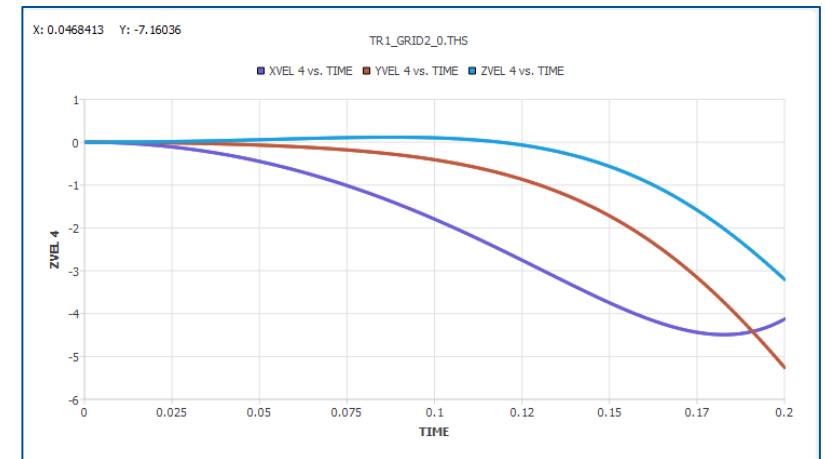
- Greater flexibility and precision in postprocessing with time-history plots

Who needs it:

- All users using time-history files when specific coordinate systems should be leveraged



X, Y, Z velocity evolution shown in global coordinate



Same X, Y, Z velocity evolution, but shown in local coordinate system

New capabilities and enhancements in **Dytran 2025.2**

Center of gravity calculation for Eulerian elements

Description:

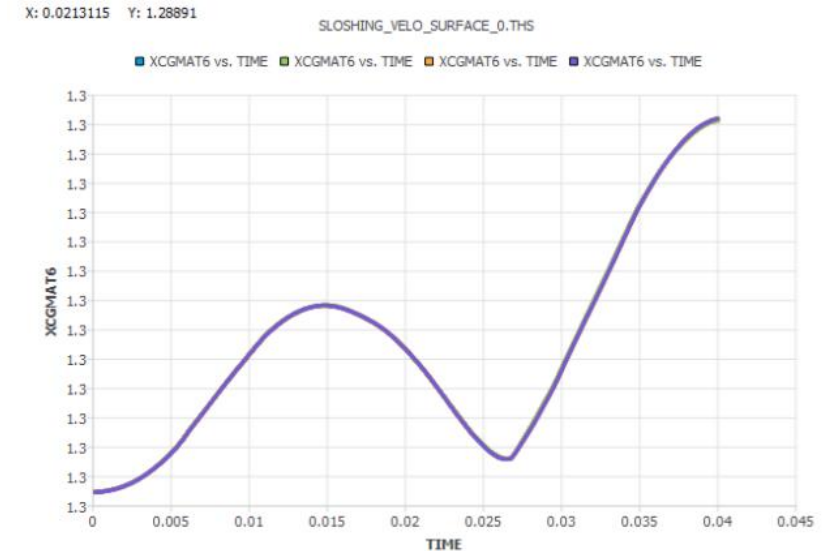
A new output capability has been introduced for fluid-structure interaction (FSI) simulations: it is now possible to extract the center of gravity for parts modeled using Eulerian elements. This enhancement is particularly valuable in scenarios where fluid displacement occurs—such as in sloshing simulations—to better assess the influence of design features like baffles. Previously, outputs such as mass, volume, and energy were available.

Benefits:

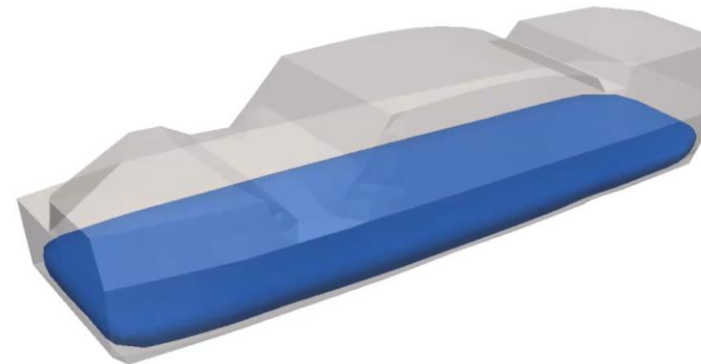
- The addition of center of gravity data provides deeper insight into fluid behavior and system dynamics.

Who needs it:

- All users performing Fluid-Structure Interaction analysis
e.g. sloshing simulation



*Sloshing analysis: x-axis motion of fluid center of gravity
(fixed Euler mesh –blue curve, adaptive Euler mesh –red curve)*



New capabilities and enhancements in **Dytran 2025.2**

New strain-based composite failure model

Description:

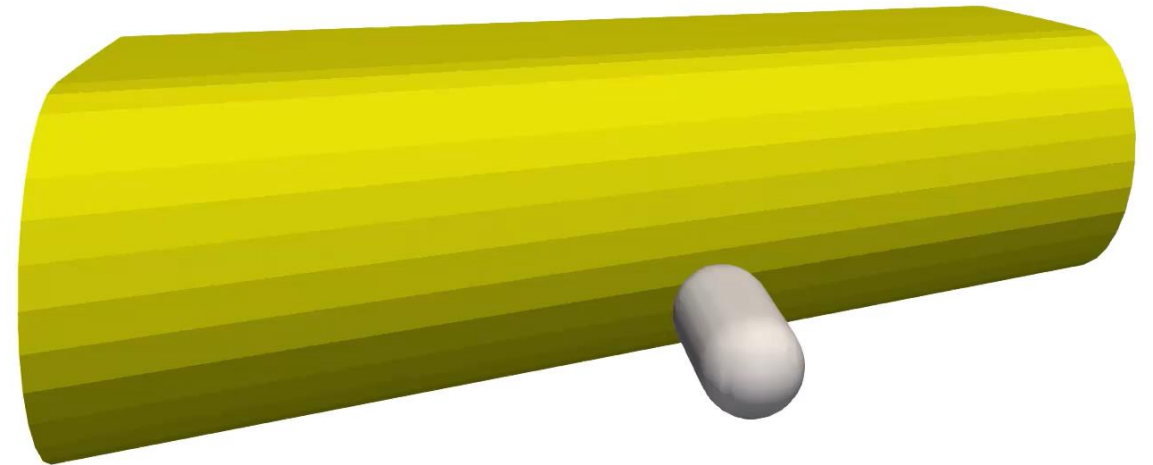
A new composite failure model has been introduced, complementing the existing stress-based models. This model introduces strain-based damage initiation, along with layer-wise damage tracking and element deletion.

Benefits:

- This new model opens the door for more accurate simulation of composite behaviour, such as in bird strike analysis.

Who needs it:

- All users performing damage and failure analysis on a composites structure



Bird strike analysis with the new strain-based composite failure model

New capabilities and enhancements in **Dytran 2025.2**

Rigid body contact improvements

Description:

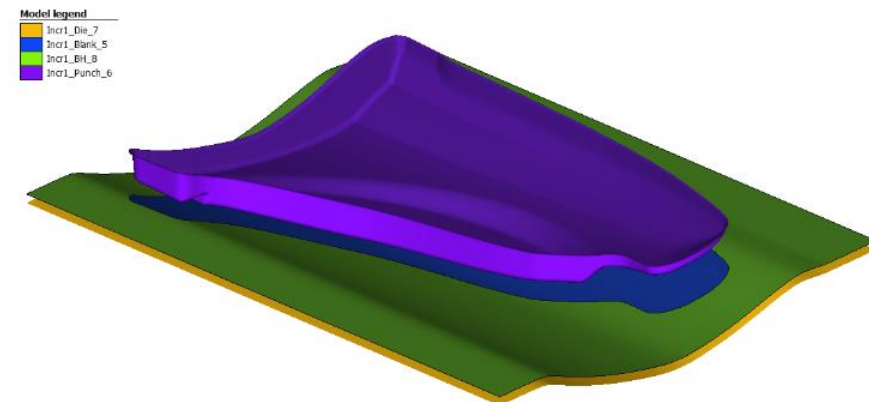
Several simulation scenarios—such as sheet metal forming—involve interactions between flexible and rigid bodies. To enhance performance in such cases, a new option has been introduced that improves parallel processing and scalability, thereby significantly reducing analysis runtime. This improvement is achieved through optimized memory management and refined surface calculations for rigid bodies.

Benefits:

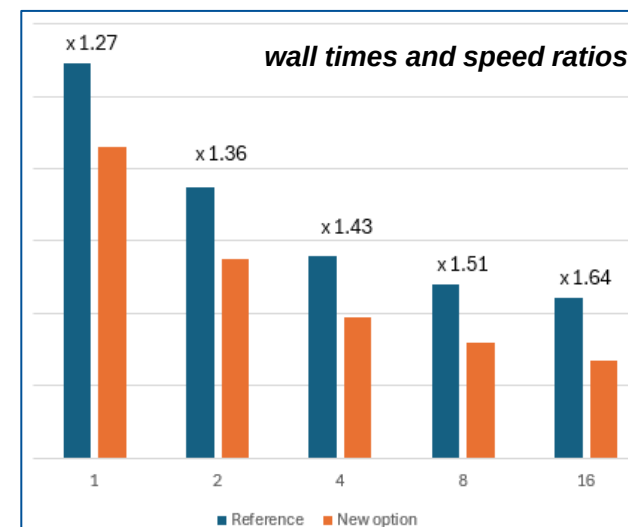
- More efficient solver performance.

Who needs it:

- All users performing interaction analysis between rigid and deformable bodies.



Sheet Metal Forming using Simufact user interface



Performance gains across CPU configurations with the new option



Previous releases

Dytran 2024.1, 2024.2 & 2025.1

Highlights from Dytran 2024.1, 2024.2 & 2025.1

Mie-Grüneisen (EOSMG) model support for Lagrangian analysis:

To model ductile materials under high-pressure, high-temperature and dynamic loading conditions.

Tied contact improvements:

New penalty-based option for tied contact between solid elements bodies.

New anisotropic plasticity models:

Two new Vegter-based material models for anisotropic plasticity have been introduced and are suitable for forming process simulation, such as rolling.

New MATEX subroutine:

New user material subroutine dedicated for Lagrangian material.

Thickness change for contact during simulation:

Contact offsets are updated at every cycle according to changes in the shell thickness.

Maximum shear stress in contact:

An option to define maximum shear stress in the friction behaviour.



Additional resources

Dytran 2025.2

Additional Resources



Support Email:
mscdytran.support@mscsoftware.com

Licensing Questions:
msc_lic.support@hexagon.com

Report a crash:
Use the Integrated Reporting Tool (IRT)

Click to visit
Nexus

Software / Dytran

Dytran

An explicit finite element analysis (FEA) solution for simulating short-duration events like impact and crash.

Contact sales

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SOFTWARE UPDATE AVAILABLE

What's New in Dytran 2022.1

Download update

01 Bush Element

02 Building UDS Libraries using oneAPI Intel Compiler

03 Modelling Large Strain Thickness Change in PCOMP CARD with Elasto-Plastic Material Model

HEXAGON

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Accurately simulate real-world physics behaviour to gain valuable insights and inform innovative designs. Test and validate designs virtually to reduce physical prototypes and improve product performance with the Hexagon Multiphysics suite. The Hexagon Multiphysics suite accelerates productivity and streamlines workflows with advanced meshing and direct modeling capabilities, easy-to-use tools for model preparation and cleanup, high-performance computing, and specialized modules for common industry vertical workflows. **Our Multiphysics Solutions:**

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Hexagon's solvers are industry-recognized for accuracy and reliability, making it possible to validate and test designs in virtual environments without relying on cost prohibitive physical prototypes. In some industries, like aerospace, Hexagon offers the only solvers trusted enough to virtually certify products for safety and durability.

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- Cadva CPD
- Actran
- CAT fatigue
- Dytran
- MSC Nastran
- Marc
- MSC Apex
- Patran
- MSC Cadw

Solutions for

- Fatigue and durability
- Lightning
- Virtual prototyping
- Wire and ribbon
- Thermal analysis
- Acoustic performance
- Fluid dynamics
- Fluid-structure interactions
- Multiphysics co-simulation

Structural simulation and analysis

Perform multidisciplinary structural analysis across static, dynamic, linear and nonlinear domains with interoperable pre- and post-processors and industry-recognized solvers.

Products

- Marc
- Patran
- MSC Nastran
- MSC Apex
- CAT fatigue

Solutions for

- Structural integrity
- Durability and fatigue
- Vibration analysis
- High-deformation scenarios

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Multiphysics
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Dytran

Explicit solution for transient structural dynamics and fluid-structure interaction

Overview

Dytran is a general-purpose, three-dimensional explicit finite element analysis (FEA) software program for simulating and analyzing complex, short-duration events involving severe deformation of structural materials and the interaction of fluids and structures. Dytran provides the most comprehensive and robust solution available to simulate impact, crush, penetration, explosion, and other fluid-structure interaction (FSI) applications. A broad range of material models and element types enable users to perform VPD experiments and "what-if" analyses on design concepts before building and testing costly physical prototypes.

Accurate, robust analysis for diverse applications

Dytran's accuracy has been proven through correlation with physical experiments. Dytran helps engineers predict how a prototype would respond to a variety of real-world dynamic events and to examine potential causes for product failure. Some examples include:

- Aerospace Applications: Aircraft ditching, fuel tank sloshing and rupture, bird strike simulation, engine blade containment, aircraft crashworthiness, seat design and safety, aircraft and cargo containment hardening.
- Automotive Applications: Airbag design and occupant safety (out of position studies), dummy modeling and seat design, vehicle impact and crash testing, tire hydroplaning, fuel tank sloshing and rupture.
- Military and Defense Applications: Shaped charge simulation and weapons design, projectile penetration and perforation of targets, hydrodynamic ram (HRAM), ship collision, underwater shock explosion (UNDEX), blast resistance and survivability.
- Other Industrial Applications: Bottle and container design, paper feeding, drop testing, sports equipment impact analysis, packaging design.

Unique combination of simulation technologies

Dytran delivers structural, material flow, and coupled FSI analyses in a single simulation package. Explicit nonlinear solver technologies are used to analyze extreme, short-duration, transient events that require finer time steps for ensuring solution accuracy and stability. This allows you to simulate FEA problems that involve (i) high degree of material nonlinearity, (ii) large geometric nonlinearity, and, (iii) extreme boundary nonlinearity.

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brochure